

Fixed Point Theorems: Conditions and Conclusions

Theorem	Domain	Mapping Type	Key Condition	Conclusion
Banach	Metric Space	Contraction	$d(Tx, Ty) \leq \alpha d(x, y), \alpha < 1$	Unique fixed point
Brouwer	$\mathbb{B}^n$	Continuous	Compact convex	Fixed point exists
Schauder	Banach	Compact map	$T(K) \subset \text{compact} \subset K$	Fixed point exists
Schauder-	Locally convex	Compact map	Compact convex set	Fixed point exists
Tychonoff	topological			
Krasnoselski	Banach	$T + S$ sum	T compact, S contraction	Fixed point exists
Goebel-Kirk	Banach	Asympt. nonexp.	Bounded convex set	Fixed point exists

Compact set property, convexity, and continuity are crucial ingredients.
Different theorems apply to different geometric and analytical settings.